

Berge–Fulkerson Covers, Cycle Double Covers, Flows and Exact Normality Defects of the First Counterexamples to the Petersen Coloring Conjecture

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CAOS Research program, github.com/fsantibanezleal/CAOS_RESEARCH

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Abstract

Jaeger’s Petersen coloring conjecture (1988), which asserted that every bridgeless cubic graph admits an edge map onto the Petersen graph sending every vertex star onto a vertex star, was refuted in August 2026: Putman exhibited two nonisomorphic 112-vertex counterexamples with SAT-solver certificates, Jooker gave a human-checkable proof, and Goedgebeur, Jooker, Máčajová, Mattiolo, Mazzuoccolo and Ulyanov found two 52-vertex counterexamples and infinite families, placing the order of a smallest counterexample between 40 and 52. The conjecture had been the strongest known sufficient condition for the Berge–Fulkerson conjecture and the 5-cycle double cover conjecture. We determine, by exact proof-carrying computation, what survives on the five retrievable counterexamples (orders 52, 52, 68, 112, 112). Independent encodings refute Petersen colorings and normal 5-edge-colorings of all five graphs with proofs checked by `drat-trim`. Every one of the five graphs admits a Berge–Fulkerson cover, a Berge cover by five perfect matchings, a Fan–Raspaul triple, a 5-cycle double cover and a nowhere-zero 5-flow, all given as explicit witnesses re-verified from the graph alone; none admits a nowhere-zero 4-flow. Their perfect matching index is exactly 4, one below the Petersen graph’s value. The two 112-vertex graphs have oddness 4 and resistance 3, whereas the other three have oddness 2 and resistance 2, every value certified from both sides. All five admit normal and strong normal 6-edge-colorings, so their normal chromatic index is exactly 6. Defining the Petersen defect as the least number of vertices at which the star condition must fail, we prove that for any edge map into the Petersen graph the number of bad vertices is never exactly one, and we find that all five graphs have defect exactly 2 with every vertex pair critical (17,362 explicit witnesses). The defect is at most the least number of abnormal edges of a proper 5-edge-coloring, and both equal 2 on the five graphs. Neither is bounded: rings of t counterexamples joined through 2-edge cuts, and cubic frames whose vertices are replaced by counterexamples minus a vertex, need one bad vertex in every block, with equality on the instances computed. Consequently no

sublinear function bounds the number of abnormal edges on 2-connected or on 3-connected cubic graphs, so four of the five statements whose equivalence was conjectured by Mattiolo, Mazzuocolo and Mkrtchyan are false. The fifth, a sublinear bound on cyclically 4-edge-connected cubic graphs, is equivalent to every such graph having Petersen defect at most 2, so the conjectured equivalence holds if and only if some cyclically 4-edge-connected cubic graph has defect at least 3. In fifteen cyclically 4-edge-connected counterexamples, the five public ones and ten new ones on 102 vertices, every pair of adjacent vertices is critical, which is consistent with the defect bound 2 on that class and against the conjectured equivalence. Finally, no counterexample consists solely of copies of the Petersen 4-pole F from which all known counterexamples are built. All computations are propositional with checked DRAT certificates or explicit witnesses. The paper decides nothing about the general covering and flow conjectures.

1 Introduction

1.1 The conjecture and its ladder

Let P denote the Petersen graph. For a cubic graph G , a *Petersen coloring* is a map $\sigma : E(G) \rightarrow E(P)$ such that for every vertex v of G there is a vertex w of P with $\sigma(\partial_G(v)) = \partial_P(w)$, where $\partial(\cdot)$ denotes the set of edges at a vertex; equivalently, the three edges at v map bijectively onto the three edges at w . A proper 5-edge-coloring is *normal* if every edge uv is *poor* ($|c(\partial u) \cup c(\partial v)| = 3$) or *rich* ($= 5$); the *normal chromatic index* $\chi'_N(G)$ is the least k admitting a normal k -edge-coloring.

Conjecture 1.1 (Jaeger 1988 [1]). *Every bridgeless cubic graph admits a Petersen coloring.*

Jaeger [2] proved that a cubic graph has a Petersen coloring iff it has a normal 5-edge-coloring (an explicit bijective form is Proposition 2.1 of [3]). The conjecture implies the Berge–Fulkerson conjecture (six perfect matchings covering every edge exactly twice) and the 5-cycle double cover conjecture [4, 9]; the Berge conjecture (perfect matching index at most 5) and the Fan–Raspaud conjecture (three perfect matchings with empty intersection) follow from Berge–Fulkerson. Every 3-edge-colorable cubic graph is trivially Petersen colorable, so the whole ladder concerns snarks. Exhaustive generation had verified the conjecture for every snark on at most 36 vertices [10] and for every weak snark on 36 vertices [11]; an exhaustive check of the weak snarks on 38 vertices shows that any counterexample has at least 40 vertices [6].

1.2 The disproof

On 6 August 2026 Putman [3] published a simple bridgeless cubic graph G_{112} on 112 vertices and 168 edges with no Petersen coloring and no normal 5-edge-coloring, certified by CaDiCaL proofs checked with `drat-trim`, together with a nonisomorphic D_3 -symmetric counterexample H_{112} ; both have girth 5 and edge- and vertex-connectivity 3. The construction composes copies of the 4-pole F (the Petersen graph minus the endpoints of one edge) with claw six-poles: $L = 4F + C$ and $G_{112} = 3L + C$. Jookan [4] gave a human-checkable proof through the Petersen-coloring sets of F and L and a 4-clique obstruction in the line graph of P . Goedgebeur, Jookan, Máčajová, Mattiolo and Mazzuocolo [5] then gave a cyclically 4-edge-connected girth-5 counterexample G_{52} on 52 vertices and infinite cyclically 4-edge-connected families. The extended account by the same authors and Ulyanov [6] gives a second counterexample G'_{52} on 52 vertices, a purely theoretical proof for both, counterexamples of every even order at least 60, and the bounds

$40 \leq n_{\min} \leq 52$ for the order of a smallest counterexample. It leaves open whether every bridgeless cubic graph has a normal 6-edge-coloring (a conjecture attributed to Šámal) and whether cyclically 5-edge-connected counterexamples exist. A further counterexample G_{68} on 68 vertices, credited there to an announcement on a social network, is available with the other four from the House of Graphs [8]. For these five graphs [6] reports Berge–Fulkerson covers, perfect matching index at most 4, 5-cycle double covers, strong normal 6-edge-colorings, and proper 5-edge-colorings with exactly two abnormal edges; the corresponding statements below were obtained independently.

1.3 What this paper does

The five counterexamples G_{112} , H_{112} , G_{52} , G'_{52} , G_{68} are the first bridgeless cubic graphs for which the Petersen route to a Berge–Fulkerson cover or a 5-cycle double cover is closed. We therefore ask, for each of them, which of the conjectures below the top of the ladder still hold, and how far each graph is from being colorable. Every answer is exact: a positive answer is an explicit witness re-verified by a standard-library checker that reads only the graph, and a negative answer is a DRAT proof checked by `drat-trim`. Section 2 gives the independent certification (a replication of the published refutations), Section 3 the perfect matching covers, Section 4 the cycle double covers, flows, oddness and resistance, Section 5 the normal 6-edge-colorings, the exact defects, and graphs of arbitrarily large defect with their consequence for sublinear approximations of the conjecture, Section 6 the compositions of copies of the Petersen 4-pole, and Section 7 the open questions.

2 Independent certification

2.1 The graphs

We use Putman’s normalized digest: the SHA-256 of the compact JSON serialization of the sorted edge list with $u < v$. The five graphs are G_{112} (digest `dc16cc18...e8b`), H_{112} (digest `0f2d8858...35c7`), both transcribed from Putman’s CC0 archive; G_{52} , transcribed from the appendix of [5] with vertices renumbered from 1..52 to 0..51 (digest `27db5d3b...`); and G'_{52} (digest `d30a423a...1477`) and G_{68} (digest `19c7c22a...5ab2`), taken from entries 57278 and 57280 of the House of Graphs [8]. Entries 57237, 57279 and 57244 of that database are isomorphic to G_{112} , H_{112} and G_{52} . The full edge lists are in the repository.

Proposition 2.1 (machine-verified). *Each of the five graphs is simple, cubic, connected, has edge connectivity 3, girth 5, and cyclic edge connectivity exactly 4: no set of at most three edges is cycle-separating (exhaustive search over all edge subsets of size at most three), and an explicit cycle-separating 4-cut exists (edges $\{2, 6, 9, 12\}$ of the sorted edge lists of both 112-vertex graphs, $\{2, 3, 8, 14\}$ of G_{52} , $\{0, 11, 70, 71\}$ of G'_{52} and $\{2, 11, 90, 97\}$ of G_{68}). The automorphism groups of G_{112} and G_{68} are trivial; those of H_{112} , G_{52} and G'_{52} have order 6.*

2.2 Encodings

Our Petersen encoding uses only edge-image variables $y_{e,f}$ ($e \in E(G)$, $f \in E(P)$), one image per edge, with the constraint that adjacent edges of G receive distinct adjacent edges of P . This suffices because P is triangle-free: three pairwise adjacent distinct edges of a triangle-free graph form a star. Two units break symmetry, using the edge-transitivity of P and the transitivity

of an edge stabilizer on the four edges adjacent to a fixed edge. Our normal-5 encoding uses edge-color variables $x_{e,c}$, side-presence variables $p_{e,c}, q_{e,c}$ (“color c appears among the two other edges at u ”, respectively at v), and a rich indicator r_e ; richness is $\neg(p_{e,c} \wedge q_{e,c})$ for every c and poorness is $p_{e,c} \leftrightarrow q_{e,c}$ for every c . Neither encoding shares a variable scheme with Putman’s vertex-witness and missing-pair encodings.

Theorem 2.2 (machine-verified, replication). *None of the five graphs admits a Petersen coloring or a normal 5-edge-coloring. All ten refutations, and the Petersen refutations of G_{112} , H_{112} , G_{52} with symmetry breaking removed, carry DRAT proofs accepted by `drat-trim` (Table 1). The same encoders accept the Petersen graph, K_4 , the prism and the flower snarks J_5 , J_7 with checker-validated colorings. Putman’s four archived proofs are accepted by our `drat-trim` binary against his archived formulas.*

instance	variables	clauses	solve (s)	proof (bytes)	check (s)
G_{112} Petersen	2,520	73,250	65.6	98,426,474	43.7
G_{112} normal-5	2,688	11,088	1,216.8	786,818,036	900.5
H_{112} Petersen	2,520	73,250	29.5	95,632,373	24.5
H_{112} normal-5	2,688	11,088	977.2	868,156,862	1,293.5
G_{52} Petersen	1,170	34,010	1.8	4,094,201	1.2
G_{52} normal-5	1,248	5,148	178.8	226,490,830	231.8
G'_{52} Petersen	1,170	34,010	13.1	4,038,980	5.0
G'_{52} normal-5	1,248	5,148	1,977.8	177,769,817	1,476.4
G_{68} Petersen	1,530	44,474	18.9	9,916,665	10.3
G_{68} normal-5	1,632	6,732	3,588.7	345,057,921	4,492.7

Table 1: Certified refutations (CaDiCaL 1.7.3 under WSL, `drat-trim`); every check printed **s** VERIFIED. The last four rows were timed while other computations shared the machine.

3 Perfect matching covers

Theorem 3.1 (machine-verified). *Each of the five graphs admits a Berge–Fulkerson cover, a Berge cover by five perfect matchings, three perfect matchings with empty intersection, and a cover by four perfect matchings. Consequently their perfect matching index equals 4.*

Proof. The matchings are explicit (the repository files of computations EXP-002 and EXP-008 list them as edge index sets) and were re-verified by a checker that tests each set to be a perfect matching and then counts the coverage of every edge. The index is at least 4 because the graphs are not 3-edge-colorable (Theorem 2.2 and the trivial direction of the conjecture). $\square$

The value 4 is one below the Petersen graph’s own perfect matching index 5, which our control recovers with a verified refutation of the four-matching instance. Thus, on the first counterexamples, the Berge–Fulkerson, Berge and Fan–Raspaud conjectures all hold, and the graphs are in this sense better behaved than the Petersen graph.

4 Cycle double covers, flows, oddness and resistance

Theorem 4.1 (machine-verified). *Each of the five graphs admits a 5-cycle double cover (five even subgraphs covering every edge exactly twice) and a nowhere-zero $\mathbb{Z}_5$ -flow, and none admits*

a nowhere-zero $\mathbb{Z}_4$ -flow. The oddness and resistance are

$$\text{odd}(G_{112}) = \text{odd}(H_{112}) = 4, \quad \text{res}(G_{112}) = \text{res}(H_{112}) = 3,$$

$$\text{odd} = \text{res} = 2 \quad \text{for } G_{52}, G'_{52} \text{ and } G_{68}.$$

Proof. Double covers and flows are explicit witnesses; the 4-flow refutations carry checked proofs. Oddness is decided from both sides: for a bound b , the formula asks for a perfect matching M and a vertex 2-coloring of the 2-factor $E \setminus M$ with at most b monochromatic edges. An odd cycle forces at least one monochromatic edge and an even cycle admits none, so the least satisfiable b is the oddness. Bounds 1, 2, 3 are refuted with checked proofs for the 112-vertex graphs and bound 4 is satisfied by a perfect matching whose 2-factor has exactly four odd cycles (counted by the checker); for G_{52} , G'_{52} and G_{68} bound 1 is refuted and bound 2 satisfied. Resistance uses a deletion indicator per edge and a proper 3-edge-coloring of the remaining edges, with the same two-sided protocol. $\square$

The oddness 4 of the 112-vertex graphs is consistent with the theorem that the smallest snarks with oddness at least 4 and cyclic connectivity 4 have order 44 [11]. The two 52-vertex graphs and the 68-vertex graph, by contrast, have the least possible oddness of a snark.

5 Normal 6-edge-colorings and the Petersen defect

5.1 Normal chromatic index

Mazzuocolo and Mkrtchyan proved that every simple cubic graph admits a normal 7-edge-coloring and asked whether 6 colors always suffice for bridgeless cubic graphs [9]; the question is restated as a conjecture in [5, 6], where strong normal 6-edge-colorings (every edge rich) of the five graphs are reported.

Theorem 5.1 (machine-verified). *Each of the five graphs admits a normal 6-edge-coloring and a strong normal 6-edge-coloring. Since none admits a normal 5-edge-coloring (Theorem 2.2), their normal chromatic index is exactly 6.*

The colorings are explicit witnesses re-verified by a checker that recomputes, for every edge, the number of colors on its two end stars. All ten instances solved within two seconds.

5.2 How far from colorable: the Petersen defect

Define the *Petersen defect* of a cubic graph G as the least number of vertices at which the star condition fails, over all maps $E(G) \rightarrow E(P)$ (equivalently, the least number of vertices whose star constraints must be dropped for a coloring to exist); a Petersen coloring is a map of defect 0. Encoding the defect through a cardinality constraint over all vertices proved hard for the solver (computation EXP-004): the bound-1 instances of G_{112} and H_{112} reached a 30-minute cap without a decision, and the bound-1 instance of G_{52} was refuted in 532.4 seconds, with a verified proof. We therefore decided the defect by *designated relaxation*: for each vertex v , a formula whose star constraints are dropped at v only; and for each pair $\{u, v\}$, a formula relaxed at both.

Theorem 5.2 (machine-verified). *The Petersen defect of G_{52} is exactly 2. For every one of its 52 vertices, the map relaxed at that vertex alone does not exist (DRAT proofs checked by `drat-trim`, each in about two seconds), while every one of the 1,326 vertex pairs admits a map whose star condition fails exactly at the two relaxed vertices (explicit witnesses; the checker reports exactly those two bad vertices; the slowest instance took 0.73 seconds).*

Thus the obstruction in the smallest known counterexample is not localized: no single vertex, in particular none of the four vertices outside the six disjoint copies of F (measured in Section 6), can absorb it, whereas every pair can. The first half is not an accident of G_{52} :

Theorem 5.3 (derived). *Let G be a cubic graph and $\sigma : E(G) \rightarrow E(P)$ any map. Then the number of vertices of G at which σ fails the star condition is never exactly one. Hence the Petersen defect of a cubic graph is 0 or at least 2, and G_{52} attains the minimum possible defect of a counterexample.*

Proof. For a vertex v let $\chi_v \in \mathbb{F}_2^{E(P)}$ be the indicator of the multiset of the three labels at v (repeated labels cancel). Every edge of G contributes its label to both endpoints, so $\sum_v \chi_v = 0$. If u satisfies the star condition then χ_u is the indicator of a vertex star of P , an edge cut. Suppose v is the only bad vertex; then $\chi_v = \sum_{u \neq v} \chi_u$ is a sum of cuts, hence a cut of P , of odd weight 1 or 3. The Petersen graph is bridgeless, so the weight is 3 and the three labels at v are distinct and form a 3-edge cut $\partial(S)$. Since P is cyclically 5-edge-connected, one side of the cut, say S , induces a forest with $c \geq 1$ components, so $3 = |\partial(S)| = 3|S| - 2(|S| - c) = |S| + 2c \geq |S| + 2$ and $|S| = 1$: the cut is a star, and v satisfies the star condition after all, a contradiction. $\square$

The argument gives more: for any map, the sum of χ_v over the bad vertices lies in the cut space of P . For the 112-vertex graphs the theorem already gives defect at least 2; whether it equals 2 is decided by pair relaxations (computation EXP-006) for the pairs of vertices outside the twelve disjoint copies of F : all 120 such pairs of G_{112} and all 120 of H_{112} are critical (explicit witnesses, at most 3.1 seconds each), so both 112-vertex graphs have Petersen defect exactly 2 as well. The full sweeps show that, as for G_{52} , every one of their 6,216 vertex pairs is critical (explicit witnesses, slowest 20 seconds), and the same holds for every one of the 1,326 pairs of G'_{52} and of the 2,278 pairs of G_{68} (computation EXP-008). The checker reports exactly the two relaxed vertices as bad in all 17,362 pair witnesses of the five graphs. We call this property, verified here for the five graphs and unexplained by the parity argument, *universal 2-criticality*. In every one of the 1,326 witnesses of G_{52} the vectors χ_u, χ_v of the two bad vertices lie in the same nonzero class modulo the cut space of P , as the argument above requires; in 1,279 of them this is the class of a single edge of P .

5.3 Abnormal edges and sublinear approximations

For a proper 5-edge-coloring c of a cubic graph G , an edge is *abnormal* if it is neither poor nor rich, and $\text{ab}(G)$ denotes the least number of abnormal edges over all proper 5-edge-colorings of G ; write $\text{pd}(G)$ for the Petersen defect. Mattiolo, Mazzuoccolo and Mkrtchyan [7] proved that no proper 5-edge-coloring of a cubic graph has exactly one abnormal edge and asked whether $\text{ab}(G) \leq 2$ implies that G has a normal 5-edge-coloring; the five counterexamples have colorings with exactly two abnormal edges and answer the question in the negative [6]. The two defects are related as follows.

Lemma 5.4 (derived). *$\text{pd}(G) \leq \text{ab}(G)$ for every cubic graph G .*

Proof. View P as the Kneser graph on the 2-subsets of $\{1, \dots, 5\}$: two subsets are adjacent when disjoint, and the *color* of the edge $\{A, B\}$ is the element outside $A \cup B$, so that the three edges at A carry the three colors outside A . Let c be a proper 5-edge-coloring with set of abnormal edges N , and for a vertex v let A_v be the complement of $c(\partial(v))$. For an edge $e = uv$, the *view from u* is the edge of P at A_u of color $c(e)$. If e is poor then $A_u = A_v$; if e is rich then A_u and A_v are disjoint and avoid $c(e)$; in both cases the two views coincide. Let $\sigma(e)$ be the common view if e is normal and the view from one chosen end if e is abnormal. A vertex all of whose edges carry its own view satisfies the star condition, so every bad vertex is the end not chosen of an abnormal edge, and there are at most $|N|$ of them. $\square$

If c has a single abnormal edge uv and the view from u is chosen, then v is bad: the view from u is an edge at A_v only if $A_u = A_v$ or $A_u \cap A_v = \emptyset$, that is, only if uv is poor or rich. The map σ would then have exactly one bad vertex, so Theorem 5.3 implies the proposition of [7] quoted above. In particular $\text{ab}(G) \geq \text{pd}(G) \geq 2$ for every cubic graph without a Petersen coloring. All five counterexamples have explicit proper 5-edge-colorings with exactly two abnormal edges (computations EXP-006 and EXP-008), hence

$$\text{pd} = \text{ab} = 2 \quad \text{for each of } G_{112}, H_{112}, G_{52}, G'_{52}, G_{68}.$$

For G_{52} the lower bound $\text{ab} \geq 2$ was also confirmed directly: the 42 single-edge relaxations refuted with checked proofs meet all 14 edge orbits of its automorphism group of order 6.

Neither defect is bounded. The constructions are those of [7]; the argument through the cut space of P gives a stronger conclusion than the one drawn there.

Theorem 5.5 (derived). *Let $G_1, \dots, G_t$, $t \geq 2$, be cubic graphs without a Petersen coloring, $e_i = a_i b_i$ an edge of G_i , and let R be obtained from the disjoint union of the graphs $G_i - e_i$ by adding the edges $a_i b_{i+1}$, indices modulo t . Then every map $E(R) \rightarrow E(P)$ has a bad vertex in every $V(G_i)$; hence $\text{ab}(R) \geq \text{pd}(R) \geq t$. If every G_i is bridgeless, so is R .*

Proof. Suppose every vertex of $V(G_i)$ satisfies the star condition under σ . With the notation of the proof of Theorem 5.3, $\sum_{v \in V(G_i)} \chi_v$ is the sum of the indicators of the labels of the two edges leaving $V(G_i)$, because an edge inside $V(G_i)$ is counted twice. Each χ_v is the indicator of a star, so the sum is a cut of P with at most two edges, hence empty, since P is 3-edge-connected. The two leaving edges therefore carry the same label x , and giving e_i the label x turns the restriction of σ into a Petersen coloring of G_i , a contradiction. If f is an edge of $G_i - e_i$, any two vertices of $V(G_i)$ are joined in $G_i - f$ by a path, in which e_i can be replaced by the detour around the ring; the added edges lie on a cycle through all the blocks. So R is bridgeless. $\square$

Theorem 5.6 (derived). *Let K be a cubic graph with t vertices and, for each vertex x of K , let G_x be a cubic graph without a Petersen coloring and v_x a vertex of G_x . Replace every vertex x of K by $G_x - v_x$, attaching the three edges of K at x to the three neighbours of v_x . Every map from the edges of the resulting cubic graph to $E(P)$ has a bad vertex in every $V(G_x - v_x)$; hence $\text{ab} \geq \text{pd} \geq t$. If K and all G_x are 3-connected, so is the result.*

Proof. If all vertices of $V(G_x - v_x)$ satisfy the star condition, the sum of their vectors χ_v is a cut of P equal to the sum of the indicators of the labels of the three leaving edges. It has odd weight, since $|V(G_x)| - 1$ is odd. Weight 1 is impossible in a bridgeless graph, so the three labels are distinct and form a 3-edge cut of P , which is a star by the argument in the proof of Theorem 5.3. Restoring v_x with these labels on its edges gives a Petersen coloring of G_x . The connectivity statement is the usual substitution along 3-edge cuts. $\square$

In [7] it is conjectured that five statements are equivalent: (a) the Petersen coloring conjecture, and the existence of a sublinear function f such that $\text{ab}(G) \leq f(|V(G)|)$ for (b) all bridgeless, (c) all 2-connected, (d) all 3-connected, respectively (e) all cyclically 4-edge-connected cubic graphs G . It is proved there that the second statement is equivalent to the first, and that on the last three classes a sublinear bound exists if and only if ab is bounded by 5, 7 and 9 respectively. The rings of Theorem 5.5 built from G_{52} are 2-connected with $\text{ab} \geq |V|/52$, and the graphs of Theorem 5.6 built from G_{52} over 3-connected cubic graphs K are 3-connected with $\text{ab} \geq |V|/51$.

Theorem 5.7 (derived). *On the class of 2-connected cubic graphs, and on the class of 3-connected cubic graphs, there is no sublinear function f with $\text{ab}(G) \leq f(|V(G)|)$ for all G in the class. More precisely, on either class such a function exists if and only if every graph of the class has a normal 5-edge-coloring.*

Proof. If every graph of the class has a normal 5-edge-coloring, the zero function is a bound. Otherwise let G be a graph of the class without one. The rings of t copies of G are 2-connected, and the graphs of Theorem 5.6 over the prisms with t vertices, $t \geq 6$ even, with every $G_x = G$, are 3-connected when G is; both have $\text{ab} \geq t$ and at most $t|V(G)|$ vertices, so ab is at least linear in the order along these sequences. The graph G_{52} is 3-connected without a normal 5-edge-coloring (Theorem 2.2). $\square$

Thus statements (a) to (d) of [7] are false, and their conjectured equivalence now amounts to the falsity of (e): by their Theorem 4, to the existence of a cyclically 4-edge-connected cubic graph with $\text{ab} \geq 10$. All five known counterexamples are cyclically 4-edge-connected with $\text{ab} = 2$. A much smaller defect would suffice.

Proposition 5.8 (derived). *If some cyclically 4-edge-connected cubic graph G^* has $\text{pd}(G^*) \geq 3$, then pd and ab are unbounded on the class of cyclically 4-edge-connected cubic graphs. Hence on that class either $\text{pd} \leq 2$ everywhere, or no sublinear function bounds ab .*

Proof. Let $e_1 = ab$ and $e_2 = cd$ be the end-edges of a path of length three of G^* , and let M be the 4-pole obtained from $G^* - e_1 - e_2$ by attaching a pendant edge at each of a, b, c, d . If M had a map to $E(P)$ satisfying the star condition at every vertex, then giving e_1 the label of the pendant edge at a and e_2 that of the pendant edge at c would produce a map on G^* whose only possible bad vertices are b and d , so $\text{pd}(G^*) \leq 2$. Hence M has no such map. Join t copies of $G^* - e_1 - e_2$ cyclically as in [7]; the result is cyclically 4-edge-connected [7, Proposition 2], and every map from its edges to $E(P)$ has a bad vertex in every copy, since a copy without bad vertices would restrict to a map on M . So its defect is at least t . $\square$

The dichotomy is in fact an equivalence, and it identifies statement (e) with a statement about two bad vertices.

Theorem 5.9 (derived). *Assume Proposition 2 of [7] (the cyclic joins used above are cyclically 4-edge-connected; its proof is omitted there). Then the following are equivalent:*

- (i) *some sublinear function f satisfies $\text{ab}(G) \leq f(|V(G)|)$ for every cyclically 4-edge-connected cubic graph G ;*
- (ii) *$\text{ab}(G) \leq 18$ for every cyclically 4-edge-connected cubic graph G ;*
- (iii) *$\text{pd}(G) \leq 2$ for every cyclically 4-edge-connected cubic graph G ;*

- (iv) *for every cyclically 4-edge-connected cubic graph G and every path $abcd$ in G , the 4-pole $G - ab - cd$ has a Petersen coloring.*

Proof. (iii) implies (ii). Let σ be a map whose bad vertices lie in $\{u, v\}$, and color every edge by the color of its image in the Kneser model of P (proof of Lemma 5.4). The three colors at a good vertex are distinct, and an edge with two good ends is poor or rich, because its two views coincide. Recolor the three edges at u : each has at most two colors forbidden at its other end, so distinct admissible colors can be chosen greedily; then recolor the edges at v in the same way (if uv is an edge, two and one admissible colors remain). The result is a proper 5-edge-coloring in which only the edges with an end in u , in v , or in one of their neighbors can be abnormal, at most $9 + 9$ edges. (ii) implies (i) trivially. (i) implies (iii) by Proposition 5.8. (iv) implies (iii) by the relabeling in the proof of Proposition 5.8. (iii) implies (iv): if some $G - ab - cd$ had no Petersen coloring, the cyclic join of three copies of it would be cyclically 4-edge-connected with defect at least 3. $\square$

Corollary 5.10 (derived). *Conjecture 3 of [7], the equivalence of statements (a) to (e), holds if and only if some cyclically 4-edge-connected cubic graph has Petersen defect at least 3 (under the assumption of Theorem 5.9).*

Proof. Statement (a) is false, (b) is equivalent to it by [7, Theorem 1], and (c) and (d) are false by Theorem 5.7. So the five statements are equivalent exactly when (e), which is statement (i) of Theorem 5.9, is false. $\square$

If, on the contrary, every cyclically 4-edge-connected cubic graph can be mapped into $E(P)$ with at most two bad vertices, then (e) holds with a constant bound and the conjectured equivalence fails.

The cyclic joining of copies of $G - e_1 - e_2$ used in [7] for that class does not force bad vertices by itself: for G_{52} , G'_{52} and G_{68} , every 4-pole obtained by deleting two independent edges has a Petersen coloring (one representative per orbit of edge pairs, 482, 482 and 4,947 4-poles, computation EXP-010), and the boundary labels of the colorings found are of two kinds only, as the cut space predicts: two crossed equal pairs, or the four edges adjacent to one edge of P . A 4-pole with a crossed pattern can be chained around a ring with an even number of copies, because any two edges of P are exchanged by an automorphism, so these cyclic joins are Petersen colorable. In all 73 4-poles of G_{52} that contain a fixed deleted edge, the two labels at the ends of that edge can be chosen adjacent in P . Ten dot products of G_{52} with itself, cubic graphs on 102 vertices with girth 5 and cyclic edge connectivity exactly 4, have no Petersen coloring (checked proofs) and defect exactly 2, and in each of them every one of the 153 pairs of adjacent vertices is critical (computation EXP-011). A pair $\{u, v\}$ of adjacent vertices of a counterexample G is critical exactly when the 4-pole $G - \{u, v\}$ has a Petersen coloring. So in all fifteen cyclically 4-edge-connected counterexamples examined, the five of Section 2 and these ten, every 4-pole obtained by deleting two adjacent vertices is colorable, and for G_{52} , G'_{52} and G_{68} so is every 4-pole obtained by deleting two independent edges. In the terms of Theorem 5.9 these computations are consistent with (iii), and hence with statement (e) and the failure of the conjectured equivalence; they do not decide it. A single cyclically 4-edge-connected counterexample with a non-critical pair of adjacent vertices would point the other way.

The bounds of Theorems 5.5 and 5.6 are attained on the smallest instances (computation EXP-009): for the rings R_t of t copies of G_{52} opened at one edge, $\text{pd}(R_2) = 2$ and $\text{pd}(R_3) = 3$, and for K_4 with each vertex replaced by G_{52} minus a vertex (204 vertices, 3-connected, girth 5)

the defect is exactly 4; each witness has exactly one bad vertex in every copy. For R_4 the defect is exactly 4 as well, with a witness found by relaxing one designated vertex in every copy. As a check of Theorem 5.5, twenty relaxations of R_2 at two vertices of the same copy were refuted with checked proofs.

6 Compositions of copies of the Petersen 4-pole

All known counterexamples contain the 4-pole F [5]. A packing computation (exhaustive enumeration of the 8-vertex sets with a 4-edge boundary that induce F , followed by an exact maximum disjoint packing) shows that G_{52} contains exactly six disjoint copies of F and four further vertices, and that each 112-vertex graph contains twelve disjoint copies and sixteen further vertices. This suggests the class $\mathcal{C}(k, m)$ of simple cubic bridgeless graphs made of k disjoint copies of F and m free vertices, the $4k + 3m$ semi-edges being joined by a perfect matching.

Proposition 6.1 (derived; machine-witnessed). *Every graph in $\mathcal{C}(k, 0)$ has a Petersen coloring. Hence no counterexample consists solely of copies of F .*

Proof. F admits a Petersen coloring in which its four semi-edges carry the same edge x of P (the distance-zero case of Jooker’s Lemma 2.1 [4]; an explicit such coloring is recorded in the repository). Color every copy by it, with the same x . Every join edge then carries x at both ends, and every vertex star is a star of a copy of F , hence a star of P . $\square$

The counterexample-guided search (computation EXP-005: join formula for the matching; structural check; Petersen-colorability of each candidate; sound blocking clauses learned from every coloring found, each recorded with its witness) exhausted $\mathcal{C}(5, 0)$ and $\mathcal{C}(6, 0)$ in seconds by exactly such a universal coloring, but did not converge on classes with free vertices: the 26-vertex control $\mathcal{C}(3, 2)$ stopped at a 15-minute time limit after 1,059 iterations and 29,052 learned clauses, and $\mathcal{C}(5, 2)$ (42 vertices) stopped at a two-hour time limit after 1,992 iterations and 80,832 learned clauses, both without finding a counterexample and without exhausting; $\mathcal{C}(6, 2)$ was not decided. We do not claim anything about classes the search did not exhaust; a symmetry-broken enumeration is a natural next step.

7 Scope and open questions

Nothing here bears on the general Berge–Fulkerson, cycle double cover, 5-flow or normal 6-edge-coloring conjectures: the computations concern five graphs. The questions left open by the disproof, the order of a smallest counterexample in [40, 52], cyclically 5-edge-connected counterexamples, and the normal 6-edge-coloring conjecture, remain open. Two questions arise from Section 5.3. Is there a cyclically 4-edge-connected cubic graph with Petersen defect at least 3? By Corollary 5.10 this is exactly the question whether the equivalence conjectured in [7] holds; a negative answer would give every cyclically 4-edge-connected cubic graph a map into $E(P)$ with at most two bad vertices. And is every pair of adjacent vertices of every cyclically 4-edge-connected counterexample critical, as it is in the fifteen examined here (every pair, adjacent or not, in the five graphs of Section 2)?

Data availability

The edge lists, encoders, checkers, witnesses and computation manifests are in the repository https://github.com/fsantibanezleal/CAOS_RESEARCH, under `problems/combinatorics/petersen-coloring/`. An identifier EXP-*nnn* denotes the archived computation in the folder `experiments/EXP-nnn-*` of that directory, with a deterministic runner and its artifacts with SHA-256 digests. Section 2 rests on EXP-001, Section 3 on EXP-002, Section 4 on EXP-003, Section 5 on EXP-004, EXP-006, EXP-009, EXP-010 and EXP-011, and Section 6 on EXP-005. The values for G'_{52} and G_{68} in Sections 3 to 5 come from EXP-008, and their certification in Section 2 from the first part of EXP-007.

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